

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CE)

December 2011

CE303 Computer Graphics

Date: 16.12.2011, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Define the following terms:

[04]

1. Aspect Ratio
2. Resolution
3. Vertical retrace
4. DVST

(b) What is Raster Scan Display? How does it differ from Random Scan Display?

[03]

Q - 2 (a) Derive Bresenham's line drawing algorithm for $|m| > 1$. Write the advantages of Bresenham's algorithm over DDA algorithm.

[07]

(b) Explain boundary fill algorithm. What is the difference between boundary fill algorithm and flood fill algorithm?

[04]

(c) Write down steps of scan line fill algorithm.

[03]

OR

Q - 2 (a) Write steps of midpoint circle algorithm for starting point $(r, 0)$. Given a circle radius $r = 10$, demonstrate the midpoint circle algorithm by determining positions along the circle octant in the first quadrant from $y = 0$ to $y = x$.

[07]

(b) Explain the methods to find out interior regions of standard polygons.

[04]

(c) Describe the following terms related to scan line fill algorithm.

[03]

1. Sorted edge table
2. Coherence property of a line

Q - 3 (a) Show that transformation matrix for a reflection about the line $y = x$, is equivalent to a reflection relative to the x axis followed by 2 counter clockwise rotations of 90° .

[06]

(b) Write a short note on homogeneous and normalized coordinates.

[04]

(c) Find the transformation matrix that transforms the given square ABCD to half its size with center still remaining at the same position. The coordinates of the square are A (1, 1) B (3, 1) C (3, 3) D (1, 3) and center at (2, 2).

[04]

OR

- Q - 3 (a) Show that two successive reflections about any line passing through the coordinate origins equivalent to a single rotation about the origin. [06]
- (b) Explain inverse transformation. Derive the matrix for inverse transformation. [04]
- (c) Apply the shearing transformation to square with [04]
A (0, 0) B (1, 0) C (1, 1) and D (0, 1) as given below
- a) Shear parameter value of 0.5 relative to the line $y_{\text{ref}} = -1$
- b) Shear parameter value of 0.5 relative to the line $x_{\text{ref}} = -1$

SECTION – II

- Q - 4 Describe in brief the following viewing parameters. What role do they play in 3D viewing process? [07]
- a) View reference point
- b) Line of sight
- c) Clipping planes
- Q - 5 (a) Use Liang-Barsky Line Clipping to clip line P1 (-1, -2), P2 (2, 4) against a window having $XL = 0$, $XR = 1$, $YB = 0$, $YT = 1$. [07]
- (b) What is polygon clipping? Explain the Sutherland-Hodgeman algorithm to clip a polygon. [07]

OR

- Q - 5 (a) Use Cohen-Sutherland out code algorithm to clip two lines P1 (40, 15), P2 (75, 45) and P3 (70, 20), P4 (100, 10) against a window A (50, 10) B (80, 10) C (80, 40) and D (50, 40). [07]
- (b) What is viewing transformation? Distinguish between viewport and window. Derive two-dimensional viewing transformation. [07]
- Q - 6 Write a short note on any TWO. [14]
- a) Back Face Detection algorithm
- b) Ambient Light Illumination model
- c) CMY Color model
- d) General 3D rotation
